

Geometry: Conditions for a Parallelogram

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DIRECTIONS

Use the 6 conditions of a parallelogram to solve for x and find missing measures. Show all work.

- 1** Opposite Sides (C2) — Solve for x :

$$AB = 3x + 5, \quad CD = 5x - 7$$

Answer: _____

- 2** Opposite Angles (C3) — Solve for x :

$$\angle A = (4x + 10)^\circ, \quad \angle C = (6x - 8)^\circ$$

Answer: _____

- 3** Consecutive Angles (C6) — Find $m\angle B$:

$$m\angle A = 112^\circ$$

Answer: _____

- 4** Diagonals Bisect (C4) — Solve for x :

$$AE = 2x + 3, \quad CE = 5x - 9$$

Answer: _____

- 5** Opposite Sides (C2) — Solve for x , find length:

$$AD = 4x - 2, \quad BC = 2x + 8$$

Answer: _____

- 6** Opposite Angles (C3) — Solve for x :

$$\angle B = (3x + 20)^\circ, \quad \angle D = (5x - 12)^\circ$$

Answer: _____

- 7** Consecutive Angles (C6) — Solve for x :

$$\angle A = (2x + 40)^\circ, \quad \angle B = (3x - 10)^\circ$$

Answer: _____

- 8** Diagonals Bisect (C4) — Solve for x :

$$BE = 3x - 1, \quad DE = 4x - 7$$

Answer: _____

- 9** Opposite Sides (C2) — Solve for x and y :

$$AB = 6x - 2, \quad CD = 4x + 6, \quad AD = y + 12, \quad BC = 3y - 4$$

Answer: _____

- 10** Find $m\angle B$, $m\angle C$, and $m\angle D$ given $m\angle A$:

$$m\angle A = 70^\circ$$

Answer: _____

- 11** Diagonals Bisect (C4) — Solve for x and y :

$$AE = 3x + 5, \quad CE = 5x - 3, \quad BE = 2y + 7, \quad DE = 4y - 5$$

Answer: _____

- 12** Opposite Angles (C3) — Solve for x and find $m\angle A$:

$$\angle A = (5x - 15)^\circ, \quad \angle C = (3x + 9)^\circ$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

C2: Opp. sides equal. C3: Opp. angles equal. C4: Diagonals bisect each other. C6: Consecutive angles are supplementary (sum = 180°).

1 Opposite Sides (C2) — Solve for x:

$$AB = 3x + 5, \quad CD = 5x - 7$$

$$= x = 6$$

Opposite sides are equal: $3x+5=5x-7 \rightarrow 12=2x \rightarrow x=6$.

2 Opposite Angles (C3) — Solve for x:

$$\angle A = (4x + 10)^\circ, \quad \angle C = (6x - 8)^\circ$$

$$= x = 9, \quad m\angle A = 46^\circ$$

Opposite angles are equal: $4x+10=6x-8 \rightarrow 18=2x \rightarrow x=9$.

3 Consecutive Angles (C6) — Find $m\angle B$:

$$m\angle A = 112^\circ$$

$$= m\angle B = 68^\circ$$

Consecutive angles are supplementary: $180^\circ - 112^\circ = 68^\circ$.

4 Diagonals Bisect (C4) — Solve for x:

$$AE = 2x + 3, \quad CE = 5x - 9$$

$$= x = 4, \quad AE = CE = 11$$

Diagonals bisect $\rightarrow AE=CE: 2x+3=5x-9 \rightarrow 12=3x \rightarrow x=4$.

5 Opposite Sides (C2) — Solve for x, find length:

$$AD = 4x - 2, \quad BC = 2x + 8$$

$$= x = 5, \quad AD = BC = 18$$

Opposite sides equal: $4x-2=2x+8 \rightarrow 2x=10 \rightarrow x=5$.
Length = $4(5)-2=18$.

6 Opposite Angles (C3) — Solve for x:

$$\angle B = (3x + 20)^\circ, \quad \angle D = (5x - 12)^\circ$$

$$= x = 16, \quad m\angle B = 68^\circ$$

Opposite angles equal: $3x+20=5x-12 \rightarrow 32=2x \rightarrow x=16$.

7 Consecutive Angles (C6) — Solve for x:

$$\angle A = (2x + 40)^\circ, \quad \angle B = (3x - 10)^\circ$$

$$= x = 30$$

Consecutive angles sum to 180°: $(2x+40)+(3x-10)=180 \rightarrow 5x=150 \rightarrow x=30$.

8 Diagonals Bisect (C4) — Solve for x:

$$BE = 3x - 1, \quad DE = 4x - 7$$

$$= x = 6, \quad BE = DE = 17$$

Diagonals bisect $\rightarrow BE=DE: 3x-1=4x-7 \rightarrow 6=x$. $BE=3(6)-1=17$.

9 Opposite Sides (C2) — Solve for x and y:

$$AB = 6x - 2, \quad CD = 4x + 6, \quad AD = y + 12, \quad BC = 3y - 4$$

$$= x = 4, \quad y = 8$$

$AB=CD \rightarrow 6x-2=4x+6 \rightarrow x=4$. $AD=BC \rightarrow y+12=3y-4 \rightarrow y=8$.

10 Find $m\angle B$, $m\angle C$, and $m\angle D$ given $m\angle A$:

$$m\angle A = 70^\circ$$

$$= m\angle B = 110^\circ, \quad m\angle C = 70^\circ, \quad m\angle D = 110^\circ$$

$\angle C = \angle A = 70^\circ$ (opposite). $\angle B = \angle D = 180^\circ - 70^\circ = 110^\circ$ (consecutive).

11 Diagonals Bisect (C4) — Solve for x and y:

$$AE = 3x + 5, \quad CE = 5x - 3, \quad BE = 2y + 7, \quad DE = 4y - 5$$

$$= x = 4, \quad y = 6$$

$AE=CE \rightarrow 3x+5=5x-3 \rightarrow x=4$. $BE=DE \rightarrow 2y+7=4y-5 \rightarrow y=6$.

12 Opposite Angles (C3) — Solve for x and find $m\angle A$:

$$\angle A = (5x - 15)^\circ, \quad \angle C = (3x + 9)^\circ$$

$$= x = 12, \quad m\angle A = 45^\circ$$

Opposite angles equal: $5x-15=3x+9 \rightarrow 2x=24 \rightarrow x=12$.
 $m\angle A = 5(12)-15=45^\circ$.

